

Artificial Intelligence Systems for Supporting Informal Caregivers of People Living with Alzheimer’s Disease or Related Dementias: A Systematic Review

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ABSTRACT

Informal caregivers of people living with Alzheimer’s disease or related dementias (PLWD) face challenges like obtaining personalized information and monitoring PLWD’s health. Rapid advancements in technology, especially in sophisticated and controversial areas like artificial intelligence (AI), prompted our study to assess AI’s potential and challenges in supporting the needs of informal caregivers of PLWD. Caregiving activities require dynamic, often unpredictable, and sometimes emotionally draining tasks that deal with a large amount of information. We conducted a systematic review to understand what AI technology has been developed to support informal caregivers of PLWD. We collected 920 papers from ACM Digital Library, IEEE Xplore, and PubMed. Screening and eligibility evaluation resulted in 16 papers for full-text review. We present which documented needs of informal caregivers have been explored by the existing research, and the contexts of the AI solutions including interfaces, data, and algorithms, as well as their effectiveness, challenges, and limitations.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in HCI**; • **General and reference** → **General literature**.

KEYWORDS

Artificial Intelligence, Informal caregivers, Dementia, Alzheimer’s disease, Human-centered design

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1 INTRODUCTION

Informal caregivers refer to individuals providing unpaid care for a friend or relative who requires special attention [14, 18, 60]. According to 2023 Alzheimer’s Disease Facts and Figures, more than 11 million Americans as informal caregivers provide unpaid care for people with Alzheimer’s disease or related dementias (PLWD) [8]. Alzheimer’s disease, a fatal degenerative condition caused by damage to nerve cells (neurons) in the brain, is currently incurable and has no known treatments to halt, prevent, or cure it [7, 8, 62]. It is the most common contributor to dementia, which, as an overall term for a particular group of symptoms, refers to difficulties with memory, language, problem-solving, and other thinking skills [7, 8]. However, informal caregivers often don’t have formal training, knowledge, and skills about caregiving activities, thus facing the challenges of time- and resource-consuming caregiving activities to maintain quality care for PLWD [60, 68].

A scoping review extracted 8 themes with 18 specific needs of informal caregivers of PLWD, covering the needs related to caregiving tasks, relationships with formal service providers, housing, juggling responsibilities, financial costs, personal health, family relationships, and planning [55]. Telephone-based, video-based, and computer-based information and communication technologies (ICT) [36, 41] and assistive technology, such as wearable sensors [25], timers [27], alarms [51], clocks [48], and calendars [6] have been developed to support informal caregivers of PLWD in caregiving tasks [43, 72], personal health [28, 36, 41], family relationships [40, 63], and planning [15, 28]. There is also growing interest in the CHI community to investigate informal caregivers’ perspectives and needs, such as examining the roles of informal caregivers when engaging PLWD with a videogame-based ICT system [71] or online activities [53], and supporting informal caregivers in communication and relationship enhancement with PLWD through interactive sound players [29], cooperative games [46], competition and skill games [46], communication systems with postcards [70], and conversational agents [77]. Researchers found that the responsibilities of informal caregivers can vary from moment to moment, depending on the conditions of PLWD [53]. As PLWD’s conditions

progress from mild or focal neuropsychological deficits to global deficits in multiple domains such as speech production, attention, or other challenges [77], the needs of informal caregivers of PLWD and their caregiving tasks are not discrete states or unchanging conditions [53]. This dynamic nature necessitates that informal caregivers flexibly adapt to the shifting demands of caregiving tasks [53]. Besides, cultural factors and genders of informal caregivers and care recipients affect the dynamics and complexity of the caregiving process [23]. With the integration of new technologies, for example, artificial intelligence (AI), into daily lives, which introduces new opportunities and challenges in the context of caregiver support, continuous evaluation and adaptation of new technologies are essential to fully meet the complex and changing needs of informal caregivers.

AI is defined as “a new generation of technologies capable of interacting with the environment by (a) gathering information from outside (including from natural language) or from other computer systems; (b) interpreting this information, recognizing patterns, inducing rules, or predicting events; (c) generating results, answering questions, or giving instructions to other systems; and (d) evaluating the results of their actions and improving their decision systems to achieve specific objectives” [19]. It has been applied in healthcare across a variety of scenarios [61], including health services management [1, 2, 11, 17, 22, 61], predictive medicine [32, 33, 35, 49], clinical decision-making [31, 37, 44], and patient data and diagnostics [17, 33, 56–58, 64, 73], demonstrating how the healthcare system can benefit from AI-supported functionalities, for example, heterogeneous health information analysis [1, 2, 11, 61, 69], real-time information updates [17, 22], automatic optimization [17, 22], and personalized treatment plans [2, 16, 61]. Such benefits could help informal caregivers in terms of personalized information retrieval, managing caregiving activities, and monitoring PLWD.

To understand how AI technology has been and can be better applied and designed to support informal caregivers of PLWD, this study aims to delve into existing literature to comprehensively understand both the met and unmet needs of informal caregivers of PLWD in the studies on AI solutions, paving the way for future research to address the gaps and limitations of current research. Besides, we applied a framework of informal caregivers’ needs [55] as guidance to examine existing AI solutions and specifically, to investigate the following research questions:

- RQ1: What needs of informal caregivers of PLWD have or have not been investigated by AI solutions?
- RQ2: What are the contexts of AI solutions for supporting the needs of informal caregivers of PLWD in terms of interfaces, data, and algorithms?
- RQ3: What are the effectiveness, challenges, and limitations of these AI solutions?

Our contributions include: 1) providing an overview of existing research on AI solutions for informal caregivers of PLWD, 2) informing the gaps in existing research in terms of the limitations of current research and the unaddressed needs of informal caregivers of PLWD, 3) discussing the challenges and opportunities of AI solutions in supporting informal caregivers of PLWD, and 4) providing a research agenda for future work.

2 METHODS

For the methods of this systematic review, we followed the phases as recommended by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [24, 52], shown in Figure 1: literature search, screening filter, eligibility evaluation, and analysis. We collected papers from multiple specialized databases, ACM Digital Library, IEEE Xplore, and PubMed, each focusing on specific disciplines [30]. Appendix A shows more details of the process.

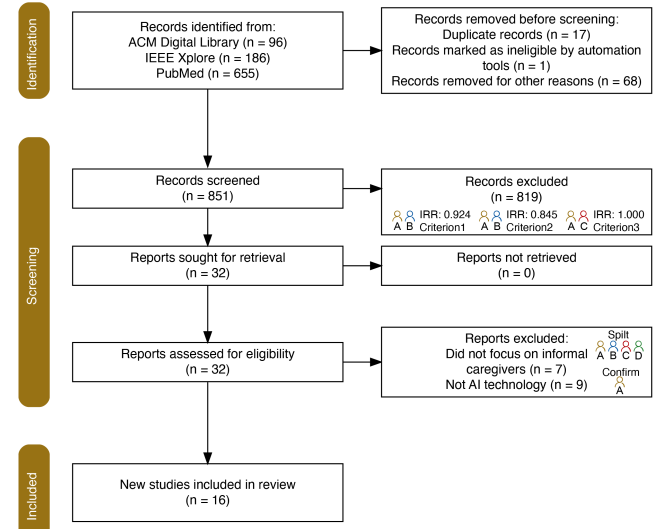


Figure 1: Diagram showing the adapted PRISMA.

2.1 Search Query

We tailored our strategies to each database. Considering the broad applicability of AI across various technologies, we incorporated solution-related keywords exclusively in the PubMed search to acquire more relevant literature. We included customer-oriented solution keywords as well, such as “telehealth” and “telemedicine,” since these solutions could also embed AI when delivering services to clients. We excluded such keywords for databases in technology, ACM Digital Library, and IEEE Xplore. This approach yielded 96 papers from ACM Digital Library, 186 from IEEE Xplore, and 655 from PubMed, encompassing literature published within the time frame from January 1st, 2013, to December 31st, 2022.

For ACM Digital Library and IEEE Xplore, our search strategy combined two components with “AND” between: (1) caregivers-related keywords, *carer* OR caregiver**, and (2) Dementia-related keywords, *dementia OR Alzheimer’s*. We also constrained the document types to be *Research Articles, Proceedings, and Journals* for ACM Digital Library and *Conferences and Journals* for IEEE Xplore. For PubMed, we applied a search query combining three components with “AND” between them: (1) caregivers-related keywords, *carer* OR informal caregiver* OR family caregiver**, (2) Dementia-related keywords, *dementia OR Alzheimer’s*, and (3) solutions-related

Table 1: An Overview of Investigated and Uninvestigated Needs based on the Framework of Needs of Caregivers of PLWD [55]

Themes	Needs	[3]	[4]	[38]	[76]	[20]	[42]	[67]	[13]	[47]	[59]	[65]	[5]	[21]	[26]	[50]	[74]	Total
1. Caregiving Tasks	1-a. Physical/nursing care						●			●								2
	1-b. Household work			●														1
	1-c. Supervision/ support	●	●				●		●		●	●	●		●		●	9
	1-d. Coordination																	0
	1-e. Help received from others (informal & formal)																	0
2. Relationship with Formal Service Providers																		0
3. Housing																		0
4. Juggling Responsibilities							●											1
5. Financial Costs																		0
6. Personal Health	6-a. Physical health																	0
	6-b. Emotional health				●	●	●							●				4
7. Relationships	7-a. With care recipient						●											1
	7-b. With family						●											1
8. Planning	8-a. Crises planning																	0
	8-b. Future planning							●										1
	8-c. Information about dementia and dementia care		●	●														2
	8-d. Information about professional support and formal services															●		1
	8-e. Information about legal regulation in caring																	0

keywords, *AI OR artificial intelligence* OR system* OR techn* OR application* OR tool* OR agent* OR telemedicine OR tele medicine OR telehealth OR tele health OR telemonitor* OR tele monitor* OR ehealth OR e-health OR mhealth OR m-health*. We constrained *Free full text and Full text* for the collection.

2.2 Screening Filter and Eligibility Evaluation

Among the 937 total papers from all three databases, we filtered out 86 papers due to duplicates, research protocols, or having no abstract, resulting in 851 papers for screening. The inclusion criteria require that the paper should focus on (1) informal caregivers as the target users; (2) Alzheimer’s disease, dementia, or mild cognitive impairments; and (3) AI-integrated solutions, including keywords related to (a) AI algorithms, such as machine learning, unsupervised learning, deep learning, recommender systems, recommendation approaches, and task-related terms (e.g., classification), (b) products known to be AI-embedded, such as Google Home, Alexa, and Siri; and (c) intelligence, for example, “smart” and “aware”.

Two researchers with research experience in informal caregiving for PLWD initially divided the 851 papers, encountering 55 overlaps for criteria (1) and (2), achieving Kappa scores [45] of 0.924 and 0.845, respectively. This process left 384 papers meeting the first two criteria. Subsequently, two researchers knowledgeable about AI split these 384 papers, noting 20 overlaps for criterion (3). After further filtration, the remaining 32 papers achieved a Kappa score of 1. These 32 papers underwent title and abstract eligibility evaluation by four researchers. Full-text reviews were then conducted to assess compliance with inclusion and exclusion criteria, with another researcher, who possesses research experience in informal caregiving for PLWD as well as knowledge of AI, verifying any papers not meeting these criteria. Ultimately, 16 papers remained after the full-text review.

2.3 Data Extraction and Analysis

To answer RQ1, we applied the framework of the needs of informal caregivers of PLWD [55]. Appendix A.4 introduces the development history of the framework and definitions of each need. To

extract the related information from the papers for RQ2 and RQ3, the research team coded two randomly selected papers together to establish consistency in the data extraction process for each code, including the PLWD’s stages, demographics of participants involved in the study, such as gender, age, and race, research goals, the informal caregivers’ needs investigated by the study, interfaces of the AI solutions, data acquired by the studies, AI algorithms, evaluation outcomes, and identified challenges and limitations. Then, four researchers equally divided the 16 papers and independently coded them. Among the four researchers, one lead researcher went through all 16 papers to confirm the validity of the codes.

3 FINDINGS

Out of the sixteen papers, seven papers (43.75%) included informal caregivers [3, 4, 20, 38, 42, 67, 76], among which two studies also involved PLWD [3, 4], two studies also involved other participants [38, 76], and the rest three studies only involved informal caregivers [20, 42, 67]. Four papers (25%) involved participants other than informal caregivers or did not inform whether the participants were informal caregivers [13, 47, 59, 65]. Five papers (31.25%) didn’t have information of participants [5, 21, 26, 50, 74], among which four papers did not involve human participants [5, 21, 50, 74], and one paper did not report the profiles of the participants [26]. PLWD’s stages and the demographics of participants were largely missing in these studies and more information is shown in Appendix B.

3.1 Needs of Informal Caregivers of PLWD Investigated and Uninvestigated by AI Solutions (RQ1)

As the Table 1 shows, of the 16 AI solutions for supporting informal caregivers of PLWD, the investigated needs were supervision/support (9, 56.25%), emotional health (4, 25.00%), physical/nursing care (2, 12.50%), information about dementia and dementia care (2, 12.50%), household work (1, 6.25%), juggling responsibilities (1,

Table 2: An Overview of Data Acquired and Interfaces Developed by the Studies

Themes	Needs	Sensors	Self-reports	Camera videos	Trusted resources	Frames of soap opera	Simulated data
1. Caregiving Tasks	1-a. Physical/nursing care	⌘ ⌘[42]		⌘[47]			
	1-b. Household work				♥[38]		
		⌘ ⌘[42]					
	1-c. Supervision/support	⌘ ⌘[3, 4, 59] ⌘[26] ⌘[13] ⌘[65]	⌘ ⌘[3, 4, 59]	⌘ ⌘[59]		NR[74]	⌘[5]
4. Juggling Responsibilities		⌘ ⌘[42]					
6. Personal Health	6-b. Emotional health	⌘ ⌘[42], NR[21]	NR[20, 21]		♥[76]		
7. Relationships	7-a. With care recipient	⌘ ⌘[42]					
	7-b. With family	⌘ ⌘[42]					
8. Planning	8-b. Future planning		NR[67]				
	8-c. Information about dementia and dementia care				♥[38, 76]		
	8-d. Information about professional support and formal services		⌘[50]		⌘[50]		

⌘ - Interfaces: Software Applications; ⌘ - Interfaces: Wearable Devices; ♥ - Interfaces: Smart Devices at Home; ⌘ - Interfaces: Smart Homes; NR - Interfaces Not Reported

6.25%), relationships with the care recipient (1, 6.25%) and with family (1, 6.25%), future planning (1, 6.25%), and information about professional support and formal services (1, 6.25%). The uninvestigated needs of informal caregivers of PLWD by AI solutions were coordination of formal services, help received from others (in-formal & formal), relationship with formal service providers, housing, financial costs, physical health, crises planning, and information about legal regulation in caring. This may stem from the difficulties in investigating the solutions with multiple stakeholders, establishing connections with relevant institutions like hospitals and legal departments, and explaining AI for decision-making.

To support informal caregivers of PLWD in supervision/support, researchers applied AI to infer and predict critical events of PLWD, such as agitation [3, 4], anxiety [59], behavioral symptoms [74], and falls [13]. Researchers also applied AI to constantly monitor PLWD [42, 65]. Helmy et al. investigated how AI can help manage PLWD's wandering behaviors by allowing informal caregivers to designate certain activities (or zones) as dangerous and triggering alert (with activity information) when trespassing was detected [26]. Andersen et al. also developed an AI-based application to involve caregivers and ordinary citizens who volunteered to locate and offer help to PLWD when PLWD got lost [5].

To support informal caregivers of PLWD in emotional health, Goins et al. applied AI algorithms to assess caregivers' burden based on survey data [20, 21]. Researchers also applied AI to provide emotional health support through smart dogs [42] and humanoid social robots [76]. Moreover, Nakazawa et al. applied AI to analyze the first-person videos recorded by caregivers and evaluated caregivers' skill levels of tender dementia-care techniques to provide physical/nursing care support [47]. Lyu et al. applied AI to help informal caregivers monitor PLWD's health conditions through wearable devices while providing companionship through smart dogs [42]. Yuan et al. developed a robot to deliver information about dementia and dementia care [76] and Li et al. applied Amazon Alexa-based devices to provide personalized education and guidance for informal caregivers of PLWD [38].

Lyu et al. found that smart dogs can also help informal caregivers of PLWD with juggling responsibilities by providing more freedom for informal caregivers, enhancing relationships with family, and relationships with care recipients [42]. Li et al. applied AI-based Alexa devices to provide recommendations to informal caregivers on food, nutrition, and cooking for PLWD [38]. Moreover, Oliva et al.

investigated applying AI to provide tailored recommendations for professional support and formal services based on user preferences and personal details, healthcare professional recommendations, and caregivers' opinions through explicit ratings given to interventions and actions such as reviewing or sharing [50]. Sunmoo et al. also used AI models to examine the relationship between demographics such as race and outcomes of social work services, thus assisting in future planning of service adoption [67].

3.2 Contexts of AI Solutions to Support the Needs of Informal Caregivers of PLWD (RQ2)

3.2.1 Interfaces developed. Among the sixteen AI solutions for informal caregivers of PLWD, four studies did not introduce the interfaces of their solutions [20, 21, 67, 74]. As Table 2 shows, there were four main types of interfaces: 1) software applications, 2) wearable devices, 3) smart devices at home, and 4) smart homes. Multiple interfaces could be applied by one study. Software applications included the digital platforms available on smartphones, tablet computers, and web browsers to provide tailored non-pharmacological educational interventions to caregivers and PLWD [50], applications to annotate or monitor symptoms of PLWD [3, 4, 42, 59], mobile app: Alzimio to monitor PLWD's activity zones [26], and the application to involve volunteers for PLWD's wandering behaviors [5]. Wearable devices were either worn by PLWD so that informal caregivers could monitor PLWD's conditions, which included smartwatches [3, 4, 42], wearable sensors [13, 59], wearable cameras [59], and stethoscope [42], or worn by informal caregivers such as wearable cameras to evaluate the caregiving skills [47]. Smart devices at home included smart dogs [42], humanoid social robot [76], and Alexa-enabled devices [38]. Smart homes referred to the one built in the Barry Lam Hall at National Taiwan University [65].

3.2.2 Data acquired. As shown in Table 2, we found researchers worked with six main types of data: sensor data (8, 50.00%), self-reported data (7, 43.75%), camera video data in real-time (2, 12.50%), data from trusted online resources including scientific literature and authorized websites (3, 18.75%), public media such as still frames of soap operas (1, 6.25%), and simulated data (1, 6.25%).

Table 3: An Overview of Algorithms and Evaluation Methods Applied by the Studies

Themes	Needs	Algorithm Types	Evaluation Methods
1. Caregiving Tasks	1-a. Physical/nursing care	OpenFace Library [47] Amazon Rekognition [47] Adaptive Network-based Fuzzy Inference System [42] CNN [47] LSTM [47] PCA [47]	■ [42] □ [47]
	1-b. Household work	Amazon Alexa [38] Android Activity Recognition API [26] LSTM [3] Axis-parallel Rectangle-based MIL [4] Boosting Bag-level Decision Stumps (MIL-Boost) [4] XGBoost Model [13] Adaptive Network-based Fuzzy Inference System [42] Multi-layered Perceptron network [59] Decision Tree [59] Random Forest [65] Finite State Machine [65] CNN [74] SVM [4, 59, 74] Dirichlet Process Mixture Model [65] Dynamic Bayesian Network [65] N-gram Model [65] NR (OpenFaas Computing Framework) [5]	■ □ [38] ▲ F [3, 4, 59] ▲ ACC [4, 13, 74] ▲ AUC [4] ▲ P [65, 74] ▲ R [65], ▲ O [5, 26] ■ [42]
	1-c. Supervision/support		
4. Juggling Responsibilities		Adaptive Network-based Fuzzy Inference System [42]	■ [42]
6. Personal Health	6-b. Emotional health	Scaled Conjugate Gradient Backpropagation Neural Network [20] Neural Network Models proposed to be used [21] Adaptive Network-based Fuzzy Inference System [42] NR (AI-embedded Humanoid Social Robot) [76]	▲ F [20] ■ [42] ■ ■ [76] ∅ [21]
7. Relationships	7-a. With care recipient	Adaptive Network-based Fuzzy Inference System [42]	■ [42]
	7-b. With family	Adaptive Network-based Fuzzy Inference System [42]	■ [42]
8. Planning	8-b. Future planning	Decision Tree [67]	▲ F [67] ▲ ACC [67] ▲ AUC [67]
	8-c. Information about dementia and dementia care	Amazon Alexa [38] NR (AI-embedded Humanoid Social Robot) [76]	■ □ [38] ■ ■ [76]
	8-d. Information about professional support and formal services	Hybrid Filtering Approach with a Content-based Algorithm and Rule-based Filtering [50]	∅ [50]

NR - Algorithm Not Reported; CNN - Convolutional Neural Network; LSTM - Long Short-term Memory Network; SVM - Support Vector Machine; PCA - Principal Component Analysis; MIL - Multiple-Instance Learning

▲ - Metric-based AI Solution Performance Evaluation | F: F-score; ACC: Accuracy; P: Precision; R: Recall; O: Other Metric Measurements; ■ - Usability Study Evaluation involving Informal Caregivers; □ - Usability Study Evaluation NOT involving Informal Caregivers; ■ □ / ■ ■ - Multiple Rounds of Usability Study Evaluation; ∅ - No Evaluation was Included

Sensor data was mainly used to support the needs of informal caregivers of PLWD in supervision/support, emotional health, relationships with the care recipient and family, and juggling responsibilities. Utilized sensor data included physiological data, such as motion, pulse, electroencephalography, cardiopulmonary condition, galvanic skin response, heart rate [13, 42, 59], body movement data captured by wearable sensors, for example, accelerometers affixed to certain specific positions of the body (e.g. waist, wrist, ankle, arm, hip) [3, 4, 13, 59, 65], location data captured by Bluetooth, gyroscope, GPS, and WiFi [13, 26, 65], and environmental sensor data of light, sound, and temperature [21].

Self-reported data was collected for supervision/support, emotional health, future planning, and information about professional support and formal service. Self-reported data captured in the studies included reports of the symptoms of PLWD such as agitation [3, 4, 21] and anxiety [59], informal caregivers' self-assessed health status [20, 21], the frequency, severity, count, and distress of caregivers caused by PLWD [20], concerns of caregivers [20], perceived support from close persons of caregivers [20], demographics [20, 67], informal caregivers' preferences, personal details, and opinions given to interventions [50], behavioral and physio-psycho-social variables, and the outcome of social work service [67].

Trusted online resources were collected to provide support for household work, emotional health, information about dementia and dementia care, and information about professional support and formal services. Such data utilized in the studies included data from scientific literature [38, 50], the United States Department of Agriculture [38], the National Heart Foundation of Australia [38], the National Institute on Aging [38], the National Institute for

Health and Clinical Excellence [50], American Association of Retired Persons [76], Family Caregiver Alliance [76], Alzheimer's Association [38, 76], Dementia.org [76], Alzheimer's.net [38], Mayo Clinic [38, 76], and Dementia Carer site [50].

Videos in real-time captured by wearable cameras can be used to evaluate the skill levels of tender dementia-care techniques for physical/nursing care [47] and to record the number of faces in the present environment for supervision/support [59]. Frames from the soap opera Emmerdale episodes focusing on the dementia storyline were applied to recognize behavioral symptoms of PLWD [74]. Simulated data of the behavior of real users who opened the app during a time window was also utilized to test the system [5].

3.2.3 Algorithms applied. Table 3 shows the algorithms applied to support the needs of informal caregivers of PLWD. These AI algorithms included established AI libraries and APIs, supervised learning, and unsupervised learning algorithms. Among the 16 AI solutions, two studies did not explicitly mention what algorithms they applied: Anderson et al. claimed that they applied OpenFaaS which included different AI techniques for online and offline anomaly detection [5] and Yuan et al. indicated their humanoid social robot applied AI [76]. Three studies applied established AI APIs or libraries: Helmy et al. utilized activity recognition API to recognize PLWD's activities [26]; Nakazawa et al. used OpenFace library and Amazon Rekognition for face recognition [47]; Li et al. applied Amazon Alexa for automatic speech recognition and natural language processing [38].

There was a diversity of algorithms applied to support the needs of informal caregivers of PLWD. Supervised learning algorithms

were applied for the classification and prediction of agitation [3, 4], anxiety [59], falls [13], and activities as well as conditions of PLWD [42, 65, 74], eye contact detection [47], and the burden of caregivers [20, 21], as well as the likelihood of resolution of issues by social work services [67]. These algorithms included neural network algorithms [3, 20, 21, 42, 47, 59, 74], support vector machine [4, 59, 74], decision trees [59, 67], axis-parallel rectangle for multiple instance learning [4], boosting bag-level decision stumps [4], XGBoost [13], finite state machine [65], and random forest [65]. Unsupervised learning algorithms were applied to analyze the data [47, 65] and provide recommendations [50], which included principal component analysis [47], Dirichlet process mixture model [65], N-gram model [65], dynamic Bayesian network [65], and a hybrid filtering approach consisting of a content-based algorithm and rule-based filtering [50].

3.3 Effectiveness, Challenges, and Limitations of AI Solutions in Supporting Informal Caregivers of PLWD (RQ3)

3.3.1 Effectiveness of existing AI solutions. For the evaluations of the effectiveness of these 16 AI solutions, shown in Table 3, two studies did not evaluate their AI solutions [21, 50]. Ten studies evaluated the performances of AI solutions based on metric measurements [3–5, 13, 20, 26, 59, 65, 67, 74]. Four studies conducted usability testing [38, 42, 47, 76], with two of them conducting multiple rounds of testing [38, 76]. Three of the four studies conducted usability testing with informal caregivers [38, 42, 76], while two of the four studies involved participants other than informal caregivers in their usability testing [38, 47].

For performance evaluations based on metric measurements, researchers reported different measurements. Researchers applied F-score to evaluate the performance of AI solutions for agitation prediction (weighted F1:0.89 [3], mean F1:0.85 [3], F-score:0.69-0.73 [4]), anxiety detection (F1:0.82-0.96 [59]), caregiver burdens assessment (F1:0.96-1 [20]), and the prediction of resolutions through medical care-related social services (F-score:0.86 [67]). Accuracy was used to evaluate the performance of AI solutions for fall detection (1 [13]), agitation prediction (0.76-0.87 [4]), activities recognition of PLWD for informal caregivers (0.315-0.524 [74]), and the prediction of resolutions through medical care-related social services (0.819 [67]). Researchers employed AUC as measurements for agitation prediction (0.78-0.92 [4]), and the prediction of resolutions through medical care-related social services (0.82 [67]). Precision (0.921-0.991 [65] or 0.296-0.546 [74]) and recall (0.975-0.995 [65]) were also applied to evaluate the AI solutions for activities recognition of PLWD. Researchers also adopted other measurements such as the confidence to achieve low delay (less than 30 seconds) of the activities detection when accuracy above 0.95 (65% confidence [26]) and the load tests for wandering issues of PLWD that the backend system was able to handle around 5000 users reliably without an increase in response time [5]. However, determining the effectiveness of these AI solutions poses a challenge, as effectiveness is highly context-dependent. Even though the metric measurements attained high scores, this does not necessarily imply the solutions effectively support informal caregivers of PLWD.

Through usability studies with informal caregivers using mixed methods, researchers received positive user experience feedback in terms of usability, readability, convenience, and accuracy of the smart dog, reduced frequency of wandering or agitation behavior of PLWD and physical discomfort during caregiving, and improved sleep quality, physical health, and the relationship with other family members and PLWD [42]. The other robot was also a high-quality user interface in terms of pleasing appearance, clear voice, appropriate gestures and movements, attractive tablet screen designs and layout, and high perceived usefulness [76]. However, participants provided a comparatively low response in terms of meeting the expectations, and one caregiver participant had doubts about using robots, believing that robots were only for PLWD [76]. Through prototype testing and data analysis, researchers proved the validity of using video-based AI systems and analysis to evaluate the skill progression of the tender dementia-care techniques [47] and investigated failed cases of dialogues with the voice assistant [38].

3.3.2 Challenges and limitations identified by researchers in using AI to support informal caregivers of PLWD. Challenges identified by researchers included 1) running sensing systems continuously without disrupting users' daily activities on affordable devices used every day (i.e., smartphones) [26]; 2) getting the ground truth due to sparsity, unpredictability, and variations in targeted behavior [4] with confounding patterns as well as inter-person and intra-person variability [3]; 3) developing standard algorithms that can adequately and concisely recognize behavioral symptoms while minimizing false alarms to keep the system reliable, usable and generalizable [21, 26, 74]; 4) dealing with the real-time processing of massive data so that activities and zones can be detected accurately in a timely fashion [13, 26]; 5) data computing including device and data calibration, data transformation, dealing with missing data and biases in subjective data [21], and integrating the information from the ambient sensors and wearable sensors simultaneously [65]; 6) recruiting older adults to participate in research studies involving technologies such as robots [76]; and 7) ethical issues when collecting or processing personal data and health information delivered to the end-user [50].

Limitations included 1) the support for informal caregivers of PLWD in late stages was still insufficient [42]; 2) difficulties for older users that it took longer for them to get accustomed to the system [42]; 3) limited accuracy that the system occasionally gave false alarms due to errors in identifying the severity of the pneumonia symptoms of PLWD [42]; 4) limited interaction capabilities such as verbal communication due to hardcoded questions and follow-up comments [76] and difficulties to match an undefined utterance to a slot [38]; 5) trade-offs in recommendation strategies including cold-start problem, sparsity problem, generalization problem, overspecialization problem, and insensitive to preference changes [50]; 6) limited generalizability of the study findings [67] including evaluation within a laboratory setting [38], small sample sizes [67, 76], short study duration [67] and limited measures and analyses [67].

4 DISCUSSION

AI outperforms ICT and assistive technologies by providing data-driven support such as assisting decision-making [31, 37, 44] and

providing personalized recommendations [2, 16, 61]. Previous studies have utilized these advantages, employing AI to support informal caregivers of PLWD in caregiving tasks such as physical/nursing care and supervision/support of PLWD, personalized support for household work, and personalized information seeking for planning. However, there are gaps in existing research.

A critical concern is the gap in understanding how the profiles of informal caregivers and the conditions of PLWD influence technology adoption and experience. According to the concept of Human-Centered Artificial Intelligence (HCAI), which emphasizes AI that not only empowers and enables human users but also openly communicates its values, biases, limitations, and ethical considerations [12], existing AI solutions for informal caregivers of PLWD, while possessing advantages over ICT and assistive technologies, often fall short in terms of transparency and ethical engagement. This becomes evident due to the largely missing demographics of informal caregivers of PLWD and the conditions of PLWD in existing research.

A lack of user-centric design (HCD) was identified in existing research. HCD is an approach to design that prioritizes the needs, motivations, emotions, behavior, and perspectives of individuals in the development process [9]. Involving anticipated end-users in the design of AI solutions prompts inquiries such as “How do researchers determine what aspects of AI can or should be co-designed?” “How does the act of participation in AI design happen?” “What do researchers do with co-designed AI artifacts?” and “How do we facilitate collaboration between co-designers and researchers with expertise across different applications domains?” [78]

Most existing studies on AI solutions for informal caregivers of PLWD have not been tested with the actual end-users—informal caregivers of PLWD—warranting the effectiveness of existing AI solutions in meeting their needs. This gap highlights the necessity for real-world testing to adhere to the principles of HCAI. The effectiveness of AI solutions depends on the context, and the relationships between metric evaluations and the user experience remain uncertain, especially considering humans have thresholds in signal detection and discrimination [10]. Will accuracy impact significant differences in experience, or is there any threshold for the metric measurement to allow a good user experience? Moreover, the varying starting levels and trajectories of trust people place when interacting with different forms of AI [19] further complicates the matter. Understanding these variances is essential for developing effective and trustworthy AI tools.

The unaddressed complexities and requirements of caregivers’ needs present significant challenges. These needs include coordination of formal services, help received from others (informal & formal), relationship with formal service providers, housing, financial costs, physical health, crises planning, and information about legal regulation in caring. Unlike supervision/support, which embeds AI concealed from informal caregivers to monitor PLWD’s condition, some of these unaddressed needs necessitate highly explainable AI solutions that are transparent and comprehensible to caregivers, particularly in collaborative decision-making domains such as housing and financial planning. Opportunities for progress and innovation may arise from gaining a deeper understanding of these needs and involving designers, developers, and informal caregivers of PLWD for AI experience design [66, 78].

5 LIMITATIONS AND FUTURE WORK

Due to the rapid expansion of research in this field, studies published since January 2023 were not incorporated. Using the identical search query, we retrieved 18 records from ACM, 45 records from IEEE, and 11 records from PubMed, published between January 2023 and March 2024. We conducted a preliminary assessment and found that these studies were in line with our findings. We will conduct a comprehensive analysis of these papers in subsequent phases of our research.

To better support informal caregivers of PLWD with their needs, future research should focus on designing and developing more caregiver-centered AI solutions. This involves a deeper understanding of the nuanced needs of caregivers across different demographics and PLWD at various stages, exploring the yet uninvestigated needs of informal caregivers, involving caregivers in the design and evaluation process of AI solutions, overcoming the limitations of current AI applications, addressing identified challenges, and ensuring that they not only meet practical needs but also resonate ethically and empathically with their users.

6 CONCLUSION

This study conducted a systematic review to explore the investigated and uninvestigated needs of informal caregivers of PLWD by AI solutions. The investigated needs were supervision/support, emotional health, physical/nursing care, information about dementia and dementia care, household work, juggling responsibilities, relationships with the care recipient and with family, future planning, and information about professional support and formal services. The uninvestigated needs of informal caregivers of PLWD by AI solutions were coordination of formal services, help received from others (in-formal & formal), relationship with formal service providers, housing, financial costs, physical health, crises planning, and information about legal regulation in caring. We examined the solution design in terms of interfaces, data, and algorithms concerning the investigated needs. We also presented the effectiveness, challenges, and limitations of meeting informal caregivers’ needs. Future research should consider the dynamics of informal caregivers’ needs and develop more caregiver-centered AI solutions.

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A MORE DETAILS OF RESEARCH METHODS

A.1 Databases

For the databases applied in this study, ACM Digital Library provides access to more than 3 million publications in computing [39] while IEEE Xplore has more than 6 million documents in electrical engineering, computer science, and electronics [75]. PubMed comprises more than 36 million citations for biomedical literature, facilitating searching across several U.S. National Library of Medicine literature resources including MEDLINE, PubMed Central, and Bookshelf [54]. These three databases will provide related literature from technology-oriented work to PLWD caregivers-oriented work.

A.2 Data collection

To improve the search strategy, the asterisk symbol “*” within a search query functions as a wildcard character, allowing matching an unlimited number of characters within a search term. This facilitates a search with broader and more adaptable outcomes. Among the 937 total papers from all three databases, we filtered out 17 duplicates, 68 research protocols, and 1 paper that did not have an abstract. The remaining 851 papers were evaluated for screening.

A.3 Inclusion and exclusion criteria

We discussed the inclusion and exclusion criteria among the research team through multiple trials of eligibility evaluation of 20

random samples per trial among two researchers to screen the titles and abstracts. Based on the inclusion criteria for screening, 819 papers were excluded and 32 papers were identified for the eligibility evaluation of titles and abstracts. The excluded examples include studies that only support PLWD, those in which caregivers are involved in the development phase but positioning PLWD as the target users, studies discussing other diseases, such as kidney disease, or diseases with risks of causing dementia (e.g. stroke and Parkinson's disease), and papers without specific AI-related keywords, such as assistive technology, internet-based supportive interventions, and tablet-based technology.

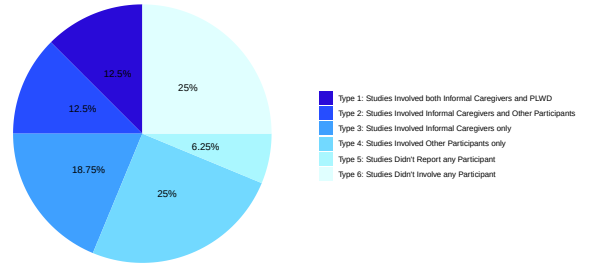
A.4 The framework of the needs of informal caregivers of PLWD

For the framework applied in this study as guidance to examine the AI solutions, researchers investigated the needs of informal caregivers of PLWD in a review of 31 articles and got 18 specific needs covering 8 themes [55] by building upon the C.A.R.E. (Caregivers' Aspirations, Realities, and Expectations) Tool, a psycho-social assessment tool. To develop the C.A.R.E. Tool, researchers identified and understood caregiver's situations including their problems, strengths, and needs, and examined the tool in terms of interrater reliability, content, and construct validity through focus groups [34]. Table A1 shows the updated 18 specific needs from the 8 themes with definitions retrieved from the original work [34] and the review [55], including physical/nursing care, household work, supervision/support, coordination of services, help received from others (informal & formal), relationship with formal service providers, housing, juggling responsibilities, financial costs, physical health, emotional health, relationship with the care recipient, relationship with family, crises planning, future planning, information about dementia and dementia care, information about professional support and formal services, and information about legal regulation in caring [55].

B PARTICIPANTS INVOLVED IN THE STUDIES

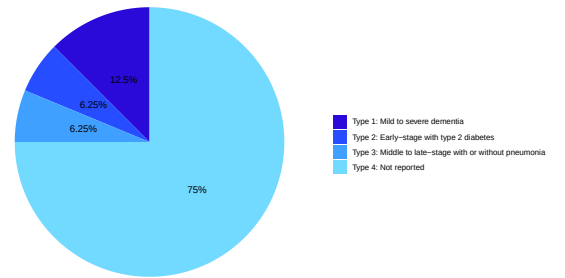
Although these studies aimed to support informal caregivers of PLWD, some of them did not involve informal caregivers to evaluate their solutions. As Figure B1a shows, among the 16 AI solutions to support informal caregivers, 4 studies (25% of the studies) did not involve participants at all [5, 21, 50, 74] because they were mainly in the stage of architecture designing. These studies presented the architectures of the AI solutions [5, 21, 50] and tested the system through established datasets [74]. One study (6.25% of the studies) did not report participants due to no direct contact with participants but the system was tested in a test bed and the field for several months using systematic sets of scenarios of usages such as users' on-foot, in-vehicle, on-bicycle, and still activities [26]. For the rest 11 studies, 2 studies (12.5% of the studies) involved both informal caregivers and PLWD [3, 4]; 2 studies (12.5% of the studies) involved both informal caregivers and other participants [38, 76]; 3 studies (18.75% of the studies) involved only informal caregivers [20, 42, 67]; 4 studies (25% of the studies) involved participants other than informal caregivers and PLWD or without clearly informing whether the participants were informal caregivers [13, 47, 59, 65].

Overview of Participants Involved in the Studies



(a) The overview of participants involved in the studies.

Overview of Stages of PLWD Involved in the Studies



(b) The overview of stages of PLWD involved in the studies.

Figure B1: Pie charts showing the overview of participants and stages of PLWD involved in the studies.

As Figure B1b shows, 75% of the studies did not report the stages of PLWD. Two studies (12.5% of the studies) focused on mild to severe dementia [3, 4]. Two studies (12.5% of the studies) reported PLWD with other diseases, thereby adding complexity to their caregiving situations. One study focused on PLWD in the early stage with type 2 diabetes [38] and the other focused on PLWD in the middle to the late stage with or without having suffered from pneumonia [42].

As Table B1 shows, existing studies varied in sample sizes with missing demographics. The sample size ranged from 1 to 607. One study using survey methodology did not report the sample size at all, only informing the percentage of different responses [42]. The most frequently reported participant demographics in the studies were gender and age, each accounting for 31.25% of the studies. Female participants were generally more than male participants [13, 38, 67, 76]. The age of the participants ranged from 25 to 85+. Three studies (18.75% of the studies) reported races, consisting of White [20, 76], African American [20, 76], and Hispanic [67]. Three studies (18.75% of the studies) reported the careers of participants such as faculty, graduate students, former paid in-home caregivers, and humanitide caregivers of different levels of experts. One study reported the education of participants and one study reported the income of participants.

Table A1: A Framework of the Needs of Informal Caregivers of PLWD

Themes	Needs	Definitions
1. Caregiving Tasks	1-a. Physical/nursing care	Informal caregivers provide physical/nursing caregiving work and support.
	1-b. Household work	Informal caregivers do work in or around the caregiving environment (home).
	1-c. Supervision/support	Informal caregivers remind/monitor/help PLWD to do the tasks.
	1-d. Coordination	Formal services coordinate with each other.
	1-e. Help received from others (informal & formal)	Informal & formal parties provide help with the caregiving tasks
2. Relationship with Formal Service Providers		Informal caregivers feel comfortable asking the medical personnel, that is hospital staff or (PLWD's) personal doctor, for information, or for more information.
3. Housing		Informal caregivers have concerns about the condition of their house as it relates to caregiving.
4. Juggling Responsibilities		Informal caregivers manage to juggle their responsibilities, commitments, and caring for PLWD.
5. Financial Costs		The financial cost of care and problems.
6. Personal Health	6-a. Physical health	Informal caregivers' specific medical health conditions.
	6-b. Emotional health	Informal caregivers' specific mental health conditions or feelings.
7. Relationships	7-a. With care recipient	The areas of tension between informal caregivers and PLWD regarding the care informal caregivers provide.
	7-b. With family	The relationships between informal caregivers and family members other than PLWD.
8. Planning	8-a. Crises planning	Informal caregivers have any plans in place for crises.
	8-b. Future planning	Informal caregivers have any plans in place for the future care of PLWD.
	8-c. Information about dementia and dementia care	Informal caregivers have needs in information about dementia and dementia care.
	8-d. Information about professional support and formal services	Informal caregivers have needs in information about professional support and formal services
	8-e. Information about legal regulation in caring	Informal caregivers have needs in information about legal regulation in caring.

Table B1: An Overview of Demographics of Participants Involved in the Study

Sample size (n)		Stakeholders	Demographics						References
Total N	Subgroup size		Gender (F/M)	Age(Mean±SD)	Race	Education	Income	Career	
16	8	PLWD	NA	F(80.5±1.1); M (80.7±7.3)	NA	NA	NA	NA	[3]
	8	Informal CG	NA	NA	NA	NA	NA	NA	
20	10	PLWD	5/5	F(80.6±4.3); M (78.7±6.8)	NA	NA	NA	NA	[4]
	10	Informal CG	NA	NA	NA	NA	NA	NA	
6	1	Informal CG	1/0	NA	NA	NA	NA	NA	[38]
	5	Others: Research team	NA	NA	NA	NA	NA	faculty (1); graduate students (4)	
10	9	Informal CG (7) + Others: older adults (2)	5/4	Age:50-65 (1); Age: 65-75 (3); Age: 75-85 (4); Age: 85+ (1)	White(5); African American (4)	Some college or higher	NA	Doctors (2); NA (7)	[76]
	1	Others: Dementia CG	1/0	Age: 65-75	NA	NA	NA	Former paid in-home CG	
607	607	Informal CG	NA	NA	Caucasian (569); African Americans (38)	NA	NA	NA	[20]
NA	NA	Informal CG	NA	Age: 25-78	NA	NA	NA	NA	[42]
72	72	Informal CG	60/12	57.3± 11.7	Hispanic	NA	81.2% had an annual income of fewer than 40,000 USD	NA	[67]
27	22	Others: Healthy subjects	17/5	NA	NA	NA	NA	NA	[13]
	5	Others: NA	NA	NA	NA	NA	NA	NA	
12	12	Others: Humanitude CG	NA	NA	NA	NA	NA	Care experts (2); middle-level CG (7); novice CG (3)	[47]
1	1	Others: NA	NA	NA	NA	NA	NA	NA	[59]
6	6	Others: NA	NA	NA	NA	NA	NA	NA	[65]
NA	NA	NA	NA	NA	NA	NA	NA	NA	[5, 21, 26, 50, 74]

CG - Caregivers; F - Female; M - Male; NA - Not Available